A Graph-Based Control Interface for Traffic Signals on Heterogeneous Road Networks

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Abstract

We present a traffic-signal control interface in which a shared graph neural network assigns scores to individual traffic movements. Each junction converts these scores into its own variable-sized set of legal signal phases using a deterministic incidence matrix. Directed corridor nodes provide traffic context, while movement nodes represent controlled input-to-output paths through junctions. Typed mean aggregation produces one scalar per movement; phase definitions and signal timing remain outside the learned network. This makes graph size and junction-specific action count independent of the learned parameter shapes. PPO experiments evaluate the interface on unseen synthetic grid geometries, altered signal coverage, and five heterogeneous city graphs. The policies retained performance across unseen geometries within the synthetic grid family, while changes in signal coverage exposed a clear distribution shift. A single trained city-policy instance executed across all five city graphs, with heterogeneous outcomes. These results provide feasibility evidence rather than a general estimate of transfer to arbitrary road networks.

1 Introduction

Traffic-signal action spaces are local. A three-arm junction, a regular four-arm junction, and a junction with protected turns need not have the same number or meaning of phases. Consequently, a fixed output called "phase 2" does not have reusable semantics across road networks.

We present a traffic-signal control interface in which a shared graph neural network assigns scores to individual traffic movements. Each junction converts these scores into its own variable-sized set of legal signal phases using a deterministic incidence matrix. The central design choice is therefore a narrow boundary: learning prioritizes comparable movement objects, while deterministic code constructs and operates each junction's valid local choices.

The resulting system combines a reusable movement-level graph representation, deterministic construction of junction-specific action spaces, and a feasibility evaluation across synthetic and city simulations. Its variable graph and action dimensions are a structural property of the design. The empirical evaluation is organized around three research questions:

RQ1 — Synthetic transfer.
How does the trained policy behave on unseen grid sizes and aspect ratios produced by the same generator?

RQ2 — Distribution shift.
What happens when signal coverage changes?

RQ3 — City feasibility.
Can the same trained policy execute on several heterogeneous city simulations, and how variable are the observed outcomes?

Movement pressure and movement-structured control have substantial prior lineage. Max-pressure methods score compatible stages from constituent flows [1]; PressLight connects pressure to learned control [2]; and FRAP represents phase competition through movements [3]. TransferLight also maps movement representations to variable phase spaces, using a learned hierarchical phase encoder [4]. The contribution here is the transparent interface between a city-level typed graph, scalar movement scores, and conflict-derived local matrices.

Scope. This report evaluates an implementation and architectural interface rather than proposing a new reinforcement-learning algorithm. The experiments evaluate transfer within the specified simulation families and execution across heterogeneous action spaces. They do not establish general transfer across arbitrary road networks.

2 Movement-level control interface

2.1 Intuition and control objects

A *movement* is one legal controlled path from an incoming road corridor to an outgoing corridor; straight travel is a movement as well as a turn. A *phase* is a compatible set of movements that may receive green together. The controller chooses one phase per junction, rather than setting individual lamps independently.

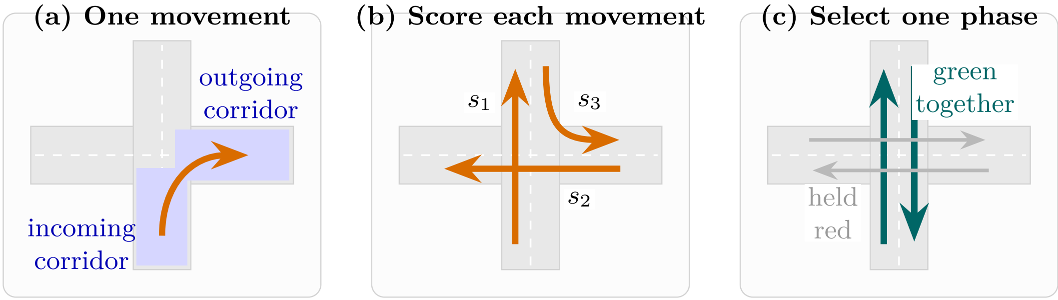

Figure 1: The control vocabulary. The shared model scores movements individually; each junction supplies its own compatible phase sets. The phase shown is illustrative, not a universal template.

The implementation groups consecutive directed road segments into a `LaneGroup` when an unsignalized continuation is unambiguous. Opposite directions remain separate because their queues, speeds, and destinations differ. At a controlled junction, a legal input-to-output connection becomes a movement node. The junction owns its movements and phases but is not itself a GNN node.

Table 1: Roles in the representation and action interface.

Object	Represents	Used for
lane group	One direction of a road corridor	Queue, speed, occupancy, capacity, arrivals, departures, and downstream space
movement	One controlled input-to-output path	Turn-local demand, controlled-link count, and recent green state
phase	One compatible local movement set	A junction-specific selectable action

At each decision, features are refreshed from SUMO, the shared GNN emits one score per movement, and a junction-local incidence matrix converts those scores to phase logits. An availability mask enforces minimum green; the runtime inserts yellow when an accepted target changes. Only movement scoring is learned.

2.2 Typed graph and message-passing architecture

Each movement m has one input LaneGroup $i(m)$ and one output LaneGroup $o(m)$, giving four directed information relations:

$$L_{\text{in}} \to M, \quad L_{\text{out}} \to M, \quad M \to L_{\text{in}}, \quad M \to L_{\text{out}}.$$

The output-to-movement relation returns downstream storage context to a movement that would feed that corridor. Unsignalized pass-through connections are weighted $L \to L$ edges with $w_{ql} = \exp(-t_{\text{ff}}/30\,\text{s})$.

Let ρ denote ReLU, $\|$ concatenation, and x_l, x_m the LaneGroup and movement feature vectors. The implementation first creates

$$h_l^{(0)} = \rho(E_L x_l), \tag{1}$$
$$h_m^{(0)} = \rho\left(E_M[x_m \| h_{i(m)}^{(0)} \| h_{o(m)}^{(0)}]\right). \tag{2}$$

For relation r, target v, and source embeddings z_q, define the transformed mean

$$\mathcal{A}_r^{(k)}(v; z) = \frac{1}{\max(1, |\mathcal{N}_r(v)|)} \sum_{q \in \mathcal{N}_r(v)} w_{qv} W_r^{(k)} z_q,$$

where $w_{qv} = 1$ except on unsignalized connector edges. The incoming and outgoing messages to movement m are

$$a_{\text{in},m}^{(k)} = \mathcal{A}_{L_{\text{in}} \to M}^{(k)}(m; h_L^{(k)}), \qquad a_{\text{out},m}^{(k)} = \mathcal{A}_{L_{\text{out}} \to M}^{(k)}(m; h_L^{(k)}). \tag{3}$$

One block first updates the movement embedding:

$$h_m^{(k+1)} = \rho\left(U_M^{(k)}[h_m^{(k)} \| a_{\text{in},m}^{(k)} \| a_{\text{out},m}^{(k)}]\right). \tag{4}$$

It then forms the return messages

$$b_{\text{in},l}^{(k)} = \mathcal{A}_{M \to L_{\text{in}}}^{(k)}(l; h_M^{(k+1)}), \qquad b_{\text{out},l}^{(k)} = \mathcal{A}_{M \to L_{\text{out}}}^{(k)}(l; h_M^{(k+1)}), \tag{5}$$
$$c_l^{(k)} = \mathcal{A}_{L \to L}^{(k)}(l; h_L^{(k)}), \tag{6}$$

and updates the LaneGroup embedding:

$$h_l^{(k+1)} = \rho\left(U_L^{(k)}[h_l^{(k)} \| (b_{\text{in},l}^{(k)} + c_l^{(k)}) \| b_{\text{out},l}^{(k)}]\right). \tag{7}$$

Every relation has its own linear map, aggregation is a mean rather than attention, and return messages use the just-updated embeddings from (4). An empty neighbourhood produces the zero vector. After two blocks, an MLP maps every $h_m^{(2)}$ to one scalar s_m. The critic mean-pools the movement embeddings owned by each junction and returns one value per controller.

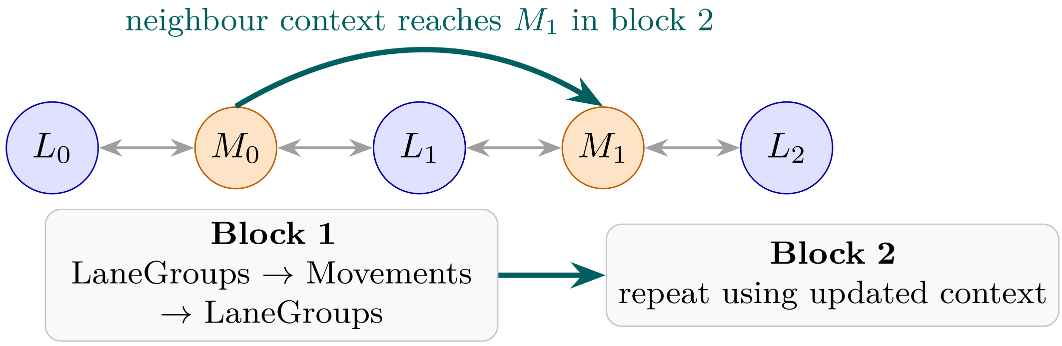

Figure 2: The typed representation and update order. With this order, two blocks allow information from one movement to reach another through a shared LaneGroup.

The parameter matrices above are shared over all nodes and edges of a type. Their shapes depend only on feature and hidden dimensions. The parameter shapes are therefore independent of graph and action-space size.

2.3 Deterministic local action spaces

The build pipeline obtains controlled links and request-conflict data from SUMO `netconvert` [5]. Links constrained to activate together form atomic groups; groups with internal SUMO conflicts are rejected. Two groups are incompatible when SUMO reports a conflict in either direction or different incoming approaches merge into the same outgoing edge. Bron–Kerbosch enumeration [6] produces all maximal compatible group sets, each of which becomes a selectable phase. Here, maximal means that no additional compatible group can be added, not that the phase has maximum size.
For junction j, $A_j \in \{0,1\}^{|P_j|\times|M_j|}$ records whether phase p enables movement m. The local logits are

$$\boldsymbol{\ell}_j = A_j \mathbf{s}_j, \qquad \ell_{j,p} = \sum_{m\in M_j} A_{j,p,m} s_m.$$

For example,

$$A_j = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \qquad \mathbf{s}_j = \begin{bmatrix} 1.2 & 0.7 & 0.6 & -0.4 \end{bmatrix}^\top, \qquad A_j \mathbf{s}_j = \begin{bmatrix} 1.9 & 0.2 \end{bmatrix}^\top.$$

The resulting logits favor the first phase; masking and categorical sampling follow before the runtime applies any signal transition. A different junction may provide a 6×11 matrix without changing the scorer.

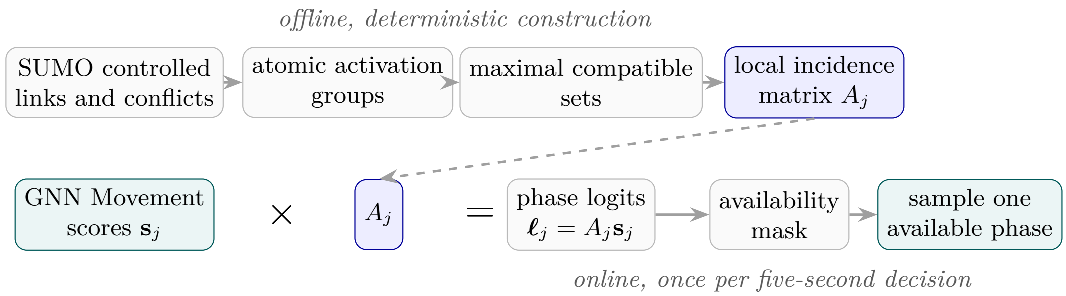

Figure 3: Offline phase construction and online action selection. The fixed incidence matrix changes with each junction; the learned scorer does not.

Sum aggregation is an additive inductive bias: when movement scores are positive, a phase containing more movements can receive a larger logit simply because it has more terms. This phase-size effect was not evaluated separately.

3 Experimental design and reproducibility

3.1 Training protocol and local reward

PPO [7] optimizes the complete policy. All reported studies use two message-passing blocks, 5 s decisions, immediate 3 s yellow on a change, one decision of minimum green, four PPO epochs, 200 decisions per rollout, and entropy coefficient 0.001. Rollout counts are allocated to approximately balance junction/action samples across differently sized training graphs.
The code assigns one reward to each junction j per decision interval $\Delta = 5\,\text{s}$. Qualitatively, the local reward encourages vehicles to move through and leave the incoming approaches while penalizing deceleration and queues. Let I_j be its unique

incoming lanes, $D_j = \sum_{l \in I_j} D_l$ their total length, n_l the vehicle count, $\bar{v}_l$ mean speed, $v_l^{\max}$ speed limit, and $V_j(t)$ the set of vehicle IDs on those lanes. With $f_l(t) = \mathrm{clip}(\bar{v}_l(t)/v_l^{\max}, 0, 1)$, the implemented terms are:

Table 2: Exact local reward terms for the reported runs. Values are sampled at decision boundaries.

Term	Definition and interpretation	Unit
Progress p_j	$D_j^{-1} \sum_{l \in I_j} n_l(t) f_l(t)$: speed-normalized moving-vehicle density	veh/m
Discharge d_j	$\lvert V_j(t-\Delta) \setminus V_j(t) \rvert / (\Delta D_j)$: IDs leaving the incoming-lane set	veh/(m s)
Braking b_j	$(\Delta D_j)^{-1} \sum_{l \in I_j} n_l(t) [\bar{v}_l(t-\Delta) - \bar{v}_l(t)]_+ / v_l^{\max}$	veh/(m s)
Gridlock q_j	$D_j^{-1} \sum_{l \in I_j} n_l(t)(1 - f_l(t))$: speed-deficit density	veh/m

The first braking sample is zero because no preceding snapshot exists. The per-junction reward is

$$r_j = \mathrm{clip}_{[-1,1]}(p_j + 10d_j - 10b_j - 0.02q_j)\,.$$

Global delay, completed-network flow, direct throughput, phase switching, and teleport terms have zero weight in these runs. Because the terms have different units and are manually weighted, the final reward is a dimensionless optimization objective.

3.2 Evaluation metrics

Every episode records per-seed metrics before arithmetic means are formed. For an episode of T simulated seconds, let C be trips with a SUMO `tripinfo` arrival after warm-up and D the initial post-warm-up population plus subsequent departures.

Table 3: Primary evaluation metrics and aggregation.

Metric	Implemented definition	Unit
Throughput	$C/(T/3600)$	veh/h
Completion	C/D; vehicles still active at the horizon remain incomplete	fraction or %
Wait density	Time mean of $\sum_l W_l(t) / \sum_l D_l$, where W_l is SUMO's total waiting time of vehicles currently on lane l	s/m

Wait density uses all non-internal network lanes at every simulated second and exposes queues left at the horizon. Grid results separate three policy-training seeds from six held-out traffic seeds.

3.3 Baseline protocol and study matrix

All policies receive the same saved SUMO network, routes, initial-occupancy range, warm-up, episode seed, conflict-derived phase set, minimum-green rule, and yellow transition. The learned and uniform-random policies choose every 5 s; fixed time cycles through phase order every 10 s; max pressure and queue recompute a target every 10 s. Max pressure observes the instantaneous halting counts in the downstream detector regions of the movement's input and output LaneGroups and scores $n_{\text{in}}^{\text{halt}} - n_{\text{out}}^{\text{halt}}$; queue observes the input detector count only. Both sum movement scores through the same A_j and retain the current phase on a tie. Fixed time and uniform random use no traffic observation. The baselines therefore do not share the learned policy's decision frequency, and comparisons include this implementation difference. The 10 s durations and all other baseline parameters were fixed in the experiment configurations; the records contain no baseline-tuning study. Max pressure is the reference for RQ1 and RQ2. For RQ3, city-specific comparisons also report the highest-throughput non-learned baseline: max pressure in Karlsruhe, queue in Mannheim, and fixed time in Stuttgart, Heidelberg, and Freiburg.

- **RQ1:** three mixed-grid training seeds, trained on five square or rectangular geometries whose longest axis is five; final evaluation on ten geometries at demands $0.6, 0.7, 0.8$, including 6×6, with six held-out traffic seeds.
- **RQ2:** a full-coverage-trained 6×6 policy evaluated after removing signals, plus one separate training seed exposed to five 50%-coverage 4×4 layouts and evaluated on held-out 25%–100% layouts.
- **RQ3:** one city policy-training seed with rollouts from Karlsruhe, Mannheim, Heidelberg, and Freiburg. Stuttgart contributed no training rollouts. Although periodic Stuttgart evaluations were available, the reported iteration-60 checkpoint was the fixed terminal checkpoint rather than one selected using Stuttgart performance. It was evaluated for 1200 s at demand scale 1.0 on three traffic seeds in all five cities.

All reported comparisons use the pinned libsumo backend.

Sampled execution is primary because it corresponds to the stochastic policy optimized during PPO training. Greedy execution performed worse in the city evaluation. The sampled results therefore do not imply equivalent performance under deterministic argmax execution.

4 Structural property and empirical results

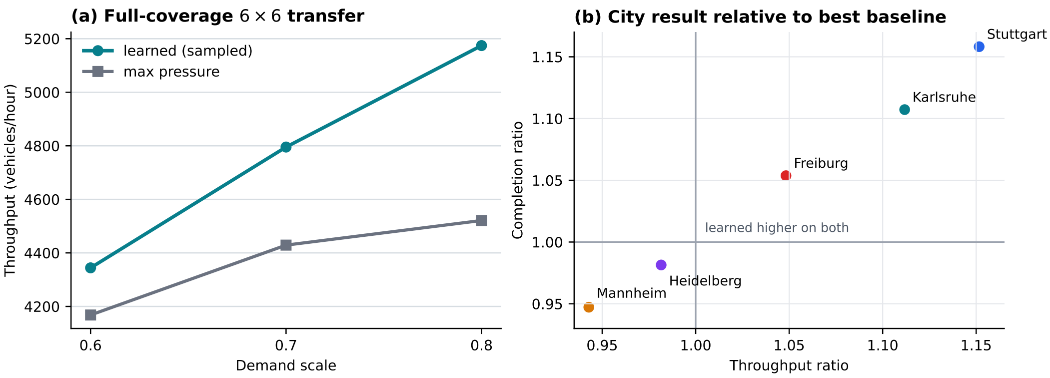

Figure 4: Observed feasibility results. Left: sampled 6×6 performance by demand. Right: iteration-60 city outcomes relative to each city's strongest non-learned throughput baseline.

Structural property — variable graph and action dimensions. Equations (4)–(7) use shared parameter shapes and return one scalar for every movement presented to them. Every junction may independently provide A_j with its own numbers of rows and columns; phase construction is outside the GNN. This establishes variable graph and action dimensions by construction. Execution on graphs ranging from 41 to 84 controlled junctions and phase-count ranges of 2–12 validates that the implementation supports the stated structural range.

RQ1 — synthetic transfer. On the unseen-size 6×6 condition, sampled throughput was $4344, 4795, 5174$ vehicles/hour at demands $0.6, 0.7, 0.8$, versus $4168, 4429, 4521$ for max pressure. Completion was higher in the evaluated runs by $3.5, 6.6, 10.6$ percentage points, respectively. At demand 0.7, the 5×2 and 2×5 cases achieved 1993 and 2005 vehicles/hour with 91.9% and 92.5% completion; the 5×3 and 3×5 cases achieved 2796 and 2791 vehicles/hour with 91.2% and 91.0% completion. These findings are consistent with size and aspect-ratio reuse within the matched generator.

RQ2 — distribution shift. The full-coverage-trained policy's throughput and completion deteriorated relative to max pressure at 50% and 25% signal coverage, despite the graph remaining structurally executable. In the separate 4×4 study, sampled throughput differed from max pressure by $+5.7$ and -28.9 vehicles/hour on held-out 25%- and 50%-coverage layouts, respectively; the differences were -104.3 and -108.0 at 75% and 100%. The result distinguishes architectural compatibility from robustness and is consistent with sensitivity to the controller-distribution shift.

RQ3 — city feasibility. The city evaluation reports one policy-training run rather than an estimate across independent training runs. A single trained parameterization executed without architecture changes on all five city graphs. Outcomes varied substantially: sampled control had higher throughput and completion than every baseline in Karlsruhe and Stuttgart, trailed queue in Mannheim, was close to fixed time in Heidelberg, and had higher throughput and completion but higher wait density than fixed time in Freiburg. Figure 4b uses the strongest non-learned throughput baseline separately for each city. The aggregate comparison instead uses max pressure consistently: for the unweighted mean across cities, the sampled policy achieved 3345.6 vehicles/hour and 64.93% completion in this run, versus 3036.8 and 59.37% for max pressure.

5 Limitations and conclusion

Limitations. The experiments evaluate the complete implementation and do not isolate individual architectural choices. They therefore do not determine the separate contributions of typed relations, two message-passing blocks, or sum-based movement-to-phase aggregation; the latter also introduces an unevaluated phase-size bias.

The evaluated networks belong to bounded synthetic and city simulation families rather than arbitrary topology. The coverage study shows sensitivity to the controller distribution, and the city study's one policy-training seed and three traffic seeds per city do not estimate variation across independent training runs.

The generated phase sets inherit the assumptions and fidelity of the SUMO network and implemented conflict rules; they are action sets accepted by the implemented construction, not formally verified real-world signal plans. Sampled execution outperformed greedy execution in the city study, so the stochastic results do not establish an equally effective deterministic controller.

Conclusion. The implemented interface separates reusable learned scoring from local signal structure: a shared typed GNN emits one scalar per movement, while deterministic incidence matrices translate those scores into heterogeneous phase spaces. That separation provides variable graph and action dimensions by construction. The experiments add bounded feasibility evidence within synthetic and city simulation families, while the coverage and greedy-execution results show that executable structure alone does not ensure empirical robustness.

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A Implementation and scenario details

Table 4: Anchored training and control settings.

Setting	Value
Decision interval / yellow	5 s / immediate 3 s
Minimum green	one decision
Message-passing blocks / hidden dimension	2 / 64
Decisions per rollout / PPO epochs	200 / 4
Entropy coefficient / reward clip	0.001 / $[-1, 1]$
Progress/discharge/braking/gridlock	1/10/10/0.02
Global/flow/throughput/switch/teleport weight	0
Initial occupancy / warm-up	6–8% / 15 s

Table 5: Structural diversity of the city scenarios. Phase range is the minimum–maximum number of selectable phases per policy-controlled junction.

City	Controllers	LaneGroups	Movements	Phase Range
Karlsruhe	41	286	271	2–10
Mannheim	84	708	467	2–7
Stuttgart	69	519	508	2–12
Heidelberg	56	408	365	2–10
Freiburg	58	425	416	2–11

B Detailed evaluation

Table 6: Full-coverage 6×6 result. Learned values average three training seeds and six held-out traffic seeds; max pressure averages the six traffic seeds.

Demand	Policy	Throughput/h	Completion	Wait density
0.6	learned sampled	4344	87.2%	0.077
	max pressure	4168	83.6%	0.082
0.7	learned sampled	4795	84.8%	0.114
	max pressure	4429	78.2%	0.119
0.8	learned sampled	5174	82.2%	0.175
	max pressure	4521	71.5%	0.131

Table 7: Iteration-60 city means over three traffic seeds. The baseline has the highest throughput among non-learned policies in that city. Wait density is in seconds per metre.

City	Policy	Throughput/h	Completion	Wait density
Karlsruhe	learned sampled	3452	93.86%	0.075
	max pressure	3105	84.77%	0.243
Mannheim	learned sampled	3200	53.78%	0.796
	queue	3394	56.78%	1.019
Stuttgart	learned sampled	4216	50.93%	0.907
	fixed time	3661	43.97%	0.741
Heidelberg	learned sampled	3190	77.69%	0.181
	fixed time	3250	79.16%	0.127
Freiburg	learned sampled	2670	48.39%	1.108
	fixed time	2547	45.92%	0.934

Sampled versus greedy. For the unweighted mean across cities, greedy execution achieved 3182.6 vehicles/hour, 61.89% completion, and 0.980 s/m wait density, compared with 3345.6, 64.93%, and 0.613 s/m for sampled execution.

Coverage study. The 50%-coverage training study used five training layouts and three held-out traffic seeds on each of five evaluation layouts at 25%, 50%, and 75% coverage; the 100% condition used one layout and the same three seeds. For sampled execution, PPO − max-pressure throughput differences at 25%, 50%, 75%, and 100% coverage were +5.7, −28.9, −104.3, and −108.0 vehicles/hour; completion differences were +0.24, −0.66, −2.99, and −3.14 percentage points.